

Exploration and Reflection on Design Research Methods in the design education in China's art academies

The preliminary exploration though World Building practice at China Academy of Art

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ABSTRACT

Taking as a case study the "World Building" course taught by the author in the Interactive Media major at the School of Animation and Games at the China Academy of Art, this article provides an initial analysis of the design education methodologies at the academy. It critiques the prevailing focus within China's design education system on "Tao" (philosophy/principles) over "Qi" (tools) and "Art" over "Technique". Additionally, it presents strategic recommendations for future reform in Chinese design education, ensuring it is adaptive to contemporary needs.

KEYWORDS

Design Research, Design Practice, Design Education in China

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1 INTRODUCTION

In the realm of international design education, world-building holds a unique and increasingly important position as a methodological tool and pedagogical approach. Originating from fields like speculative fiction, video game design, and filmmaking, world-building has transcended its original domains to become a critical aspect of design thinking and education.

World-building is not merely the construction of an alternate reality; it is a comprehensive design process that requires a multi-disciplinary approach. From the intricacies of societal structures to the details of environmental contexts, world-building allows designers to holistically understand and create complex systems. This immersive design mode fosters a wide range of skills including critical thinking, systemic thinking, narrative skills, and an ability to understand and design for complex user experiences.

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As we step further into the 21st century marked by rapid technological advancements and increasing complexities, the traditional design education models focused on specialized skill sets are often found to be inadequate. World-building equips students with the ability to see "the big picture," emphasizing the interconnectedness of various elements in a system. It enhances empathy, broadens perspectives, and encourages interdisciplinary thinking. Furthermore, it opens avenues for incorporating ethics, philosophy, sociology, and other fields into design, making the education more rounded and inclusive.

In academic settings like the Rhode Island School of Design (RISD), MIT Media Lab, and various other leading design and tech schools around the globe, world-building is being integrated into the curriculum to prepare the next generation of designers to tackle real-world problems that are increasingly systemic and interrelated. Courses often involve students creating detailed and immersive worlds as part of their design process, encouraging them to think about the implications, ethics, and complexities of their design choices. In summary, world-building is gaining recognition for its ability to foster a more holistic, research-based, and forward-looking approach to design. It helps to prepare future designers for a world that is not just increasingly interconnected but also increasingly complex, demanding a multifaceted skill set and a deeply ingrained capacity for adaptive, critical thinking. This is precisely what is currently lacking in China's design education.

2 ISSUES IN CHINESE DESIGN EDUCATION

The true initiation of modern art and design education in China began in the early 1980s, primarily originating from the transformation of the craft art disciplines. There has been continuous debate and reflection on design education research within the country. Similar to the general delay of theory compared to practice, art and design education have always been behind the actual application of design in various industries, which is more evident in contemporary Chinese art and design education. On one hand, due to rapid technological development, China's modern manufacturing industry quickly rose, with high-tech represented by information technology influencing almost all economic sectors and manufacturing industries. While the manufacturing industry achieved leapfrog development and rapidly increasing output, a disparity emerged between the output of China's higher design education and the needs of the modern industry. China's modern art and design education, which started not long ago, provided lagging services for China's modern industry with its immature design education theory and teaching practices.

2.1 Brief Introduction of Design Education in China

The Central Institute of Arts and Crafts, presently known as the Academy of Arts & Design at Tsinghua University, serves as an archetypal representation of the metamorphosis in pedagogical paradigms within the sphere of China's design education. This apex institution elucidates the transition from an initial emphasis on arts and crafts toward a more systematically structured design education framework. Historically, during the period spanning the 1950s to the 1980s, academic programs in art and design within mainland China were primarily housed within departments of Arts, Fine Arts, and Craft Arts in conventional universities. Post-1980, specialized design majors commenced emerging in universities with an engineering focus.

The curriculum content was predominantly rooted in the domains of craft arts and decorative design, encompassing sub-disciplines such as packaging design, advertising design, textile arts, and bookbinding. Additionally, the realm of industrial product design was chiefly anchored in traditional artisanal crafts, including but not limited to, ceramic design, lacquerware, and specialized crafts.

2.2 The Dilemma of Design Education in China's Art Academies

In terms of the education system, design education was long based on pure art education and traditional craft art education, forming a "heavy on art, light on technique" tendency, especially lacking in practical technology education, rigorous design research methods, and training for proactive thinking skills. This phenomenon continues today and is one of the key factors contributing to China's design education lagging behind international modern design education. Modern design education, which began in the Bauhaus School in Germany in the 1920s, has always emphasized the integration of art and technology. This requires design students to have certain research and innovation capabilities. However, this type of research on the real world has been long missing in China's art and design education, which is an issue that deserves serious reflection.

3 "RESEARCH THROUGH DESIGN" AS A NEW ATTEMPT

Amid the burgeoning complexities of the 21st-century economic landscape, China's design education system exhibits discernible lacunae in aligning itself with industrial imperatives, thereby necessitating a paradigmatic shift. Employing a "research through design" methodology could serve as a seminal venture to bridge this disconnect, fostering a more symbiotic relationship between academic pedagogy and industry needs. This innovative approach would not only bolster critical thinking and adaptive problem-solving capabilities but also facilitate a more holistic understanding of the interconnected dynamics between design, technology, and societal demands, elements that are currently underemphasized in the extant educational model.

3.1 Challenges

The "Unified Design Education Model" [See Yuan, 2003] in China has led to the Chinese design education system lagging behind once again in the economic competition of the 21st century. The successful cases of other Asian countries, such as South Korea, Singapore, and Thailand, in the synchronous development of design education and industry, are enough to prove the problems of contemporary Chinese art and design education. Today, in the 21st century, the complexity triggered by technological iteration and the cross-disciplinary nature of knowledge urgently require the Chinese design education system to break out of the "siege" for reform.

3.2 Origins

The behavior of conducting research through design is essentially observing the real world, analyzing, and summarizing, and designing simulations under preset conditions to seek potential solutions. Norman Holland argued in "Literature and the Brain" that the association cortex of the human prefrontal lobe serves this behavior, and this "simulation" of reality plays an important role in human survival and reproduction. The "simulation" behavior that humans possessed early on can be seen as an activity to create a fictitious world based on the ability to observe and research the real world. This activity is the core of design methods such as world-building, and its content is concretely manifested in mediums such as novels, films, and games.

Professor Alex McDowell of the University of Southern California first practiced the design method of world-building in "Minority Report" (2002). Although world-building methods had been used in science fiction movies before, the future technology in "Minority Report" has a forward-looking quality that elevated the design method of world-building to a new height: the touch screen technology appeared in the film has had a huge impact on later technological developments (e.g., Apple's iPad, touchscreen mobile phones, etc.). It is worth mentioning that Alex McDowell currently guides the World Building Media Lab and World Building Institute at the University of Southern California and incorporates world-building as a post-cinema narrative design practice into his teaching. At present, world-building has become a cross-disciplinary field in design, technology, and narrative storytelling.

In 2014, Alex McDowell brings World Building Media Lab at USC and Design Interactions at RCA in a joint course, marking the first intersection of world-building and Speculative Design method. [Dunne and Raby, 2013] The course revolved around the fictional city island Rilao, inspired by a 1986 peninsula invention by José Saramago, breaking from the mainland, driven by the separation of Los Angeles and Rio de Janeiro, with real-world elements serving as Rilao's "DNA." RCA students shaped Rilao, designing artifacts (products, systems, services, tools, cultural items) presented as design prototypes and films. Greek student Ira Pappa explored multi-gender culture's social impact, creating a society with unique gender identities and freedom. Taiwanese student Paul Gong focused on animals' function in human society. His Rilao combined Los Angeles humans and Rio animals, using biotechnology to design tailored animal tools, challenging human-animal-nature relationships. The course displayed 13 pieces combining real-world research

and projected social problems onto Rilao. As a student of Anthony Dunne and Fiona Raby's MA program "Design Interactions" at the Royal College of Art, I also participated in this joint course. During this course, I discovered several commonalities between the design methods of Speculative Design and World Building:

- a. Proposing questions based on research of the real world.
- b. The need for imagination that breaks away from the framework of the real world.
- c. Attempting to solve real-world problems with a fictional world and these aspects are precisely what is lacking in China's current design education.

4 "WORLD BUILDING" COURSE IN CHINA ACADEMY OF ART

Re-examining the contemporary Chinese art and design education model and system is not an easy task, as it differs significantly from the way of thinking shaped by Chinese design education. When I first came into contact with these emerging design methods in the UK, I also spent a great deal of time learning, researching, and practicing repeatedly in the design process to find my own path to understanding these design methods. Modern Chinese design and its education originate from craft art education and have long been affiliated with pure fine arts (painting). From the pattern education in the early Republic of China period to today's art design education, over a century of evolution, China's art design education has experienced different development stages from "pattern" to "craft art", and then to "art design". Although we introduced the "Bauhaus" and Western design education models and systems respectively in the Republic of China period and the early 1980s, this "introduction" did not penetrate the core of the education system, manufacturing industry, and economic system, but only stayed at the level of formal imitation, showing a state of "non-compliance with local conditions". In other words, if we just completely "introduce" Western design methods, Chinese students may experience "non-compliance with local conditions" due to differences in cultural background and way of thinking. Therefore, while addressing the current issues in Chinese design education, we must also localize the "introduced" design methods.

The "world-building" course mentioned in this chapter is a course I was offered for undergraduate third-year students, at the School of Animation and Games in China Academy of Art's last year, mainly based on a design-driven research method. The core of the world-building method lies in design research thinking, which is also its essential difference from traditional design courses. This course requires students to design and think about the overall system of a fictional world, guiding them to integrate research into the early stage of the design process and cultivate the important design ability of research into practice through the repetitive use of research materials in the design process. In this course, I primarily focus on the solutions to the following three issues:

- Proposing questions based on research of the real world.
The core problem that the students' design work ultimately solves must be extracted from the real world.
- Attempting to solve real-world problems with a fictional world.

The extracted core problem is to be solved in the world ten years from now.

- The need for imagination that breaks away from the framework of the real world.

Changes in the world ten years from now are extrapolated through vertical research on a particular technology.

Based on the above three points, I designed the following brief for this course: In the year 2032, the world's temperature has reached its highest ever. However, the pace of developing infrastructure for protection from urban heat has not kept up with the pace of temperature change. The premise is that reducing urban heat exposure by increasing access to shade is an equity issue because low-income communities are likely to live in neighborhoods with fewer trees, poor building insulation, more heat-retaining built surfaces, and limited access to cool spaces that provide respite during a heat wave. Write questions that your character (considering aspects such as body, place, society, system, culture, etc.) in the year 2032 would ask as they attempt to find probable solutions to combat the effects of heat. The questions should start with... "What if?" [Dunne and Raby, 2013] [Cechanowicz et al., 2016] and they should continue to be in the first person.

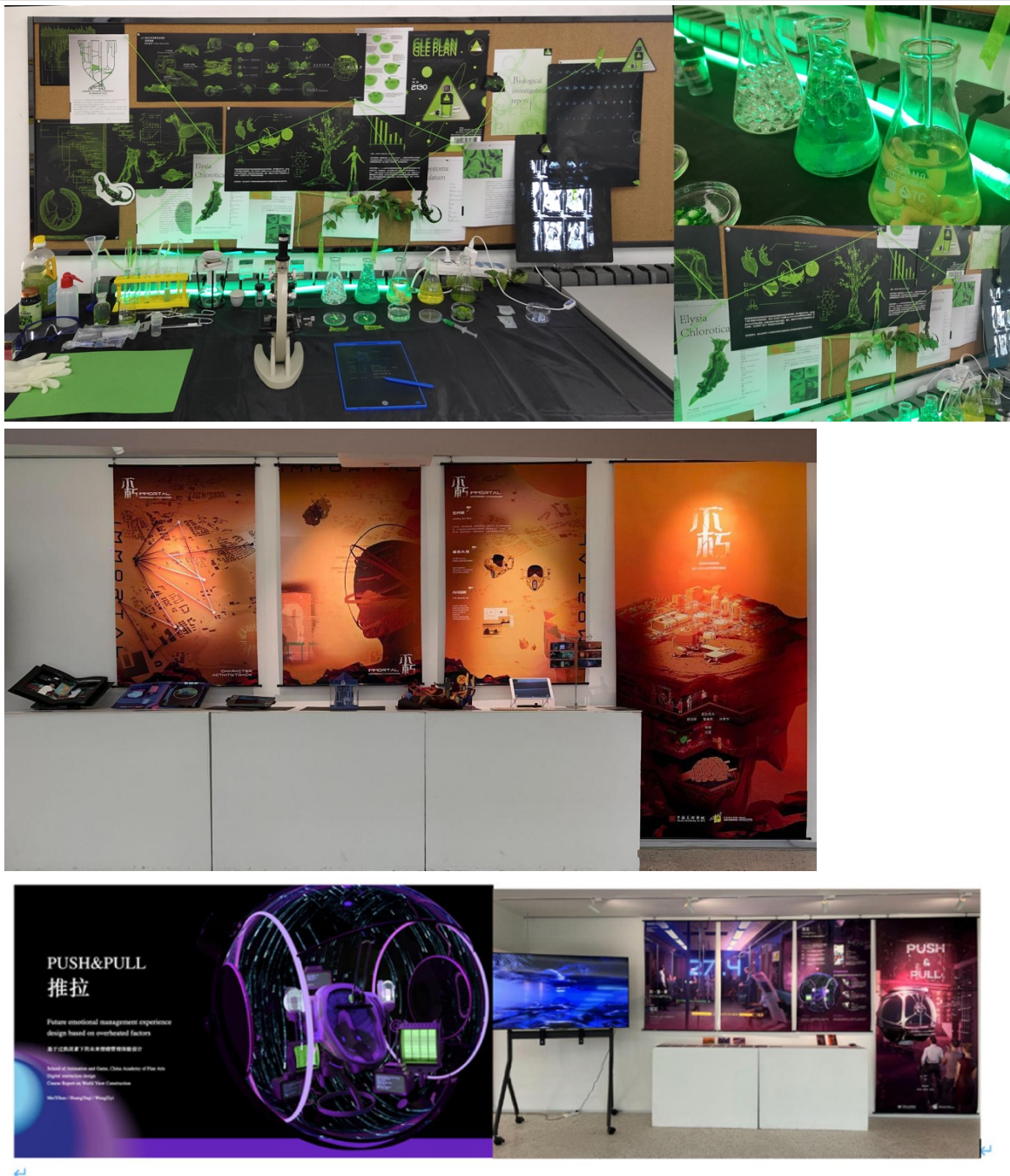
Tips:

1. The more questions you write, the more it will contribute to the extrapolation of the world in 2032.
2. Pick a place near you (or one you're familiar with) and record as much of that location as possible. Take photos. Not only shoot the space but also objects and structures; your 2032 world will be deduced based on this location.

Over a five-week period, approximately one week is spent understanding and analyzing the key points and fundamental research in the brief, including the inspection and collection of data on the real-world environment. From the second week, the deepening of the research begins. In this week's course, students will use tools (such as the McKinsey framework [McKinsey Quarterly, 2008]) to differentiate "weak signals" from mainstream trends and other disruptive events. These will be used in a clustering exercise using a mapping technique that can be applied to many different product, service, and organizational change scenarios. The third and fourth weeks begin the design prototype phase, in which students will need to repeatedly return to the research materials from the first two weeks, using the design research tools provided by the teacher to repeatedly verify the feasibility and credibility of the prototype, and produce a prototype for presentation. In the final week, students will need to review the entire design process, identifying key points or paths to choose the best way to publicly present their prototypes and research.

Partial student work sees below:

In total, over 30 students ultimately formed 7 groups of works. Each set of work includes a large poster showcasing the research path, design materials that support the narrative, physical objects created through 3D printing or material production, along with printed materials, short film and videos. At the end of the course, students were required to hold a small exhibition in an area designated by the school.



5 COURSE REFLECTION AND FUTURE PROSPECTS

In the process of conducting the world-building course, design students are strongly urged to remain vigilant about the binary opposition between fiction and reality. While the literal meaning of "design" has a clear functional attribute, from a broad design perspective, "design" seems difficult to define simply. For instance, product design primarily involves everyday items such as furniture

or tools; graphic design leans towards the application of visual language and graphic symbols, such as print materials or packaging; while interaction design is more about how to naturalize operations without the need for reading instructions.

World-building is essentially an immersive design mode: it uses design prototypes to ferment people's ideals, attitudes, and ways of viewing the world, enhances empathy, and introduces a new

dimension that stimulates imagination and interdisciplinary thinking. This model emphasizes the evolution of logical thinking and the subversion and reshaping of perspectives on things, especially when faced with the challenges of complexity or interdisciplinary knowledge brought about by the iterative technology of the 21st century. This design method is a very beneficial mental exercise, especially crucial for training design students in design research and further advancing design practice. Comparatively, these types of design methods often have strong criticism, and their core is intangible, making this kind of design sometimes elusive and even unconventional. Sometimes, from a visual perspective, digital virtuality, VR and AR, and other digital expanded experiences, science fiction, and the design of the fictional world seem very close. However, the basis of fiction is still purposeful stacking in the real world, so its uniqueness lies in the potential to evolve into an objective existence. Therefore, using design as a research medium to place the discovered problems in a space beyond reality for pre-testing, its purpose is to create a context through students' research to visualize, immerse, and make the problem discussable. In other words, through such a method that uses design as research means, we can analyze the development and change trends of the surrounding world and create a form of communication to visualize those indirectly visible or deliberately ignored issues.

Given that world-building has the characteristic of design criticism, it requires students to engage in research on real-world issues. Unlike traditional design characterised by functionalism since the 20th century, the design of world-building is a conscious way of manufacturing reality. Traditional design clearly can no longer meet the needs of the complex world of the 21st century. The key to 21st century design is understanding people: design is increasingly manifesting as a way of thinking. In areas where there are no answers, asking questions is the beginning of thinking. Its primary task is not to find solutions, but to understand through extensive research what factors have led to the problem.

Due to the divergence-dominated thinking and the lack of independent thinking abilities among the majority of students fostered by China's education system, the journey to "accept" the attitude of conducting research and to learn how to do research remains a long one in China. At present, China's design education has begun to extend to primary and secondary schools, so integrating "research training" into early-stage education might be essential. Meanwhile, by delving into these issues in the curriculum, I am currently developing a basic course in design research for lower-level undergraduates.

6 CONCLUSION

The concept of design itself continues to evolve in the course of history. In the early agrarian society, design was primarily manifested as a craft; entering the industrial society, design turned towards empowering, servicing, and being driven by industrial forms of technology. Now, the information industry and technology, especially the rapid development of artificial intelligence, have had profound effects on our future modes of social production and life. Correspondingly, new design challenges have followed. Our past understanding of design was often based on intuitive levels, hence many design considerations for the problems at that level have

arisen. However, the future design will focus more on the state of ultramicro, the intangible, the networked.

Today, as the global design methods and thought modes are changing, China's design education must make corresponding adjustments, by reforming the existing educational structure and ways to better prepare future designers for the rapidly changing societal and world development needs. Today, we are at a critical period for the reform and development of China's design education. Faced with the increasingly complex technological environment of the 21st century, designers will take on more and more important responsibilities. The global pandemic that broke out in 2020 reminds us that even if something seems impossible, it does not mean it will not happen. In this context, would the act of liberating design imagination and the design method of conducting research in advance help us rediscover and extrapolate the possibilities of the relationship between humans, society, and reality? We can set real-world problems in future world environments and analyze potential crises and potentials in future societies. We can also use design as a research method to step beyond reality, to consider the pros and cons of problems in a "dystopian" context, to make rules, detect impacts, and create new possibilities between society and reality. This kind of thinking is closely connected with the idea that Chinese design education aims to cultivate designers who are multifaceted and critical. A major challenge of our time is to reflect on what has happened in design education and to use these reflections to improve the current education system.

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